

Multi-Domain Vision Transformer Fusion for Intersectional Demographic Classification from Facial Images

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This research was supported in part by the Department of Informatics, Faculty of Informatics, Universitas Negeri Surabaya.

ABSTRACT Simultaneous recognition of facial demographic attributes such as race and gender faces challenges from expression variations, biological aging, phenotypic overlap, and the limitations of single-domain representations. This study proposes a multi-domain latent feature fusion framework that integrates visual representations from three task-specific pre-trained Vision Transformer (ViT) models for facial biometrics, affective expression, and age estimation, coupled with classical machine learning pipeline optimization for classifying six intersectional demographic subgroups on the DemogPairs dataset. Latent representations are extracted offline from each model to produce a unified visual representation that combines complementary cross-domain information. Seven feature configurations are evaluated through 5-Fold Stratified Cross-Validation across four classifiers: Random Forest (RF), Gaussian Naive Bayes (GNB), Logistic Regression (LR), and Support Vector Machine (SVM) with hyperparameter tuning via Grid Search Cross-Validation (GridSearchCV). Experimental results demonstrate that tri-domain fusion achieves the best performance across three of the four evaluated classifiers, with the SVM model yielding the highest performance, recording an Accuracy of 93.70%, Precision of 93.72%, Recall of 93.70%, and F1-Score of 93.69% on independent test data. In-depth evaluation across the six subgroups records an F1-Score range between 91.74% and 96.14%, demonstrating the effectiveness of multi-domain latent representation fusion in recognizing intersectional demographic attributes.

INDEX TERMS Race and gender classification, intersectional demographic recognition, multi-domain feature fusion, algorithmic fairness, Vision Transformer.

I. INTRODUCTION

Automated facial demographic attribute recognition plays a pivotal role across various intelligent application domains, including digital forensics systems, biometric access control, human-computer interaction, and personalized interactive services [1], [2]. However, multiple studies have reported performance disparities across specific demographic subgroups, particularly females and individuals with darker skin tones [3], [4]. Such disparities can be influenced by various factors, including data distribution imbalances and the limitations of feature representations in handling visual variations [5]. The limitations of conventional systems in coping with such visual variations demonstrate that fair and accurate demographic recognition still faces tangible practical challenges, thereby necessitating novel representation approaches capable of modeling the interaction between racial and gender demographic attributes in an integrated and comprehensive

manner.

While some facial attribute recognition approaches model attributes separately, multi-task learning enables multiple attributes to be learned simultaneously [6]. Modeling attributes in isolation can restrict the analysis of groups formed by attribute combinations, such as race and gender [7], [8]. Consequently, aggregate metrics potentially fail to capture performance disparities across intersectional demographic subgroups [9]. Controlled evaluation across demographic subgroups with balanced representation can provide a more consistent and objective comparative foundation for measuring inter-group performance variations [9]. Intersectional modeling also faces representational challenges because facial characteristics and expressions can vary across demographic groups, thereby necessitating visual representations capable of preserving discriminative information while remaining robust against such variations [10].

The methodological evolution of facial attribute recognition has shifted from handcrafted features toward deep learning paradigms, including CNN and ViT. In the era of Convolutional Neural Networks (CNN), various studies explored demographic feature extraction by leveraging specific facial regions such as the mid-face area, multi-scale multi-task frameworks for simultaneous age and gender estimation, and deep CNN models for ethnicity recognition [11], [12], [13]. ViT is capable of modeling global inter-patch relationships via Multi-Head Self-Attention (MHSA), thereby potentially yielding more contextual visual representations, while the utilization of synthetic generative representations has also been developed to mitigate demographic disparities [14], [15]. In addition, the integration of parallel attention mechanisms, the application of ViT for facial gender classification, and the exploration of multi-axis transformer architectures have further enriched the diversity of visual representation modeling [16], [17], [18]. Furthermore, exploration into combining cross-domain ViT representations has emerged through the integration of dual facial and emotional features to enhance race and gender attribute discrimination [19], [20]. Nevertheless, the fusion of biometric, expression, and aging representations into a unified representation remains relatively limited.

Despite the continuous advancements in ViT, several research gaps directly relevant to intersectional demographic classification remain. First, the majority of prior studies rely on single-domain facial biometric representations, while the simultaneous integration of affective information for intersectional demographic classification remains limited [11], [17], [18]. Second, demographic recognition frequently exhibits high sensitivity to age-related facial morphological changes because the employed feature extraction does not incorporate biological aging representations [12], [17]. Third, comparative studies regarding the influence of classifiers, feature scaling, and dimensionality reduction on transformer representations remain limited [13], [19], [20]. To bridge the limitations of single-domain representations and evaluate these downstream classifier decision boundaries, this study proposes an integration of three ViT representation domains combined with structured classical classifier pipeline optimization.

To address these research gaps, this study proposes a Tri-Domain ViT feature fusion framework that combines three task-associated latent representations for intersectional demographic classification from facial images. The proposed framework integrates three complementary representation domains, including facial biometric representations, affective expressions, and aging characteristics, into a unified representation via feature fusion [21], [22]. Latent feature extraction is conducted offline from frozen pre-trained ViT backbone models, allowing downstream classification to be performed without fine-tuning the backbone [23]. The resulting feature vectors are subsequently evaluated through single-domain, dual-domain, and tri-domain ablation schemes using four classical machine learning classifiers, namely RF, GNB, LR, and SVM. All downstream classifiers are evaluated using Stratified Cross-Validation within a pipeline that integrates

feature scaling and PCA dimensionality reduction, with each transformation learned strictly on the training data of each fold to prevent information leakage [24].

The main scientific contributions of this study are summarized into four primary points as follows:

- 1) A Tri-Domain ViT feature fusion framework integrating face-associated biometric representations, expression-related representations, and age-associated facial representations into a unified latent feature vector for intersectional demographic classification.
- 2) An empirical comparative benchmark across multiple feature ablation schemes and classical machine learning classifiers optimized via Stratified Cross-Validation hyperparameter tuning, examining decision boundary behavior across linear, probabilistic, ensemble, and kernel-based models.
- 3) Competitive classification performance on the Demog-Pairs dataset, with the proposed tri-domain representation outperforming the evaluated single-domain and dual-domain configurations for the majority of classifiers.
- 4) A granular intersectional subgroup performance analysis, evaluating subgroup-level classification performance across six demographic subgroups, providing an in-depth characterization of model behaviors across racial and gender intersections while systematically revealing the consistency of multi-domain feature representations on independent evaluation cohorts.

The remainder of this article is organized as follows: Section II discusses prior research on facial demographic recognition and transformer architectures. Section III details the DemogPairs dataset specifications, multi-domain ViT representation extraction, and the optimized classification pipeline. Section IV presents the experimental evaluation results, feature ablation studies, and intersectional subgroup performance analysis. Finally, Section V summarizes the primary research conclusions, methodological limitations, and recommendations for future research directions.

II. RELATED WORKS

Facial demographic attribute recognition has transitioned from handcrafted descriptors toward deep representations. Local region-based approaches leverage the geometry of the central facial region to recognize ethnicity and mitigate background interference [11]. However, CNN-based approaches still face challenges in maintaining performance under illumination variations and facial characteristic discrepancies across racial groups, including the Other-Race Effect phenomenon. Increasing the depth of convolutional networks can enrich visual feature representations to differentiate diverse populations [13]. In addition, parallel feature-sharing mechanisms with visual attention have been developed to capture inter-attribute correlations while mitigating local representation limits. This approach reflects a paradigm shift from feature engineering toward representation learning that is more adaptive to facial diversity [16].

The application of ViT in facial biometrics offers visual modeling capabilities distinct from convolutional approaches. Through the Multi-Head Self-Attention (MHSA) mechanism, transformers can model inter-patch relationships globally, thereby capturing broader spatial dependencies compared to local convolution operations [14]. These characteristics demonstrate the potential of transformers in modeling demographic attributes, including gender, on standardized-resolution facial images [17]. Beyond standard ViT architectures, hybrid architectures such as MaxViT combine window-based and grid-based attention to construct hierarchical spatial representations [18]. Although transformers possess powerful global representation capabilities, their performance remains influenced by the quality, scale, and distributional characteristics of the training data utilized during the learning process.

Performance disparity investigations in computer vision systems have driven evaluation protocols that account for balanced class and demographic subgroup distributions. Testing on balanced datasets provides controlled experimental conditions for comparing performance across subgroups by mitigating the confounding impact of sample size imbalances. However, distributional balance does not inherently eliminate social biases embedded within the data. Latent representation analysis further demonstrates that transformer models can reproduce social biases originating from large-scale pre-training corpora [14]. To reduce performance disparities across racial and gender groups, generative augmentation based on StyleGAN2 has been employed to synthesize variations that help mitigate classification errors [15]. These findings underscore the critical importance of systematically conducting evaluations across demographic subgroups.

The limitations of single-domain representations in capturing the diversity of facial characteristics have motivated the exploration of multi-task learning and representation fusion from multiple information sources. Multi-task approaches have been utilized to learn correlations between age and gender attributes through jointly processed facial representations [12]. Parallel representation-sharing mechanisms have also been developed to combine diverse affective attributes simultaneously, whereas generative approaches can enrich feature variations across demographic groups [16], [15]. Furthermore, the integration of ViT-based latent features uniting facial biometric representations and emotional expressions has demonstrated potential for improving intersectional attribute classification [19]. Combining biometric representations with age information also exhibits complementary informational contributions, establishing cross-domain fusion as a compelling approach [20].

Once feature representations are obtained, the choice of downstream classifier governs how high-dimensional features are mapped to the decision space. End-to-end approaches commonly employ a SoftMax layer to map feature representations directly to class probabilities [11], [18]. Alternatively, Support Vector Machines (SVM) can learn non-linear decision boundaries within the feature space without

updating the frozen backbone parameters [13]. Tree-based ensemble models such as RF and probabilistic approaches such as GNB can also serve as classifiers with distinct complexity and class-separation characteristics [12]. These differences demonstrate that classifier selection substantially impacts the effectiveness of latent representation utilization, particularly when features originate from multiple domains and exhibit high dimensionality.

Based on the reviewed literature, significant opportunities remain to explore facial demographic attribute modeling through the integration of representations across multiple domains. In contrast to prior studies that predominantly rely on single-domain representations or partial domain combinations, this study proposes a tri-domain fusion framework that unifies facial biometric geometric representations, affective expression cues, and biological aging morphological characteristics using pre-trained ViT models with frozen backbones. The latent representations from the three domains are subsequently concatenated and fed into an SVM to learn non-linear decision boundaries. Evaluations are conducted on the balanced DemogPairs dataset to benchmark intersectional race and gender classification while systematically analyzing performance disparities across all evaluated subgroups.

III. MATERIALS AND METHODS

The proposed framework integrates multi-domain visual representations and machine learning model optimization for intersectional race and gender classification, as illustrated in Figure 1. The methodology begins with partitioning 224×224 pixel resolution facial images from the DemogPairs dataset into training and test partitions via a stratified split. Three frozen pre-trained ViT-Base models extract 768-dimensional latent representations from the [CLS] token offline. The three vectors are combined through concatenation fusion $\mathbf{z}_{\text{tri}} = \mathbf{f}_{\text{face}} \oplus \mathbf{f}_{\text{emotion}} \oplus \mathbf{f}_{\text{age}}$, where $\mathbf{z}_{\text{tri}}$ denotes the 2,304-dimensional tri-domain vector, $\oplus$ is the concatenation operator, and $\mathbf{f}_{\text{face}}$, $\mathbf{f}_{\text{emotion}}$, and $\mathbf{f}_{\text{age}}$ represent facial biometric, expression, and age features, respectively. The representation vectors are processed within a modular Scaler and PCA pipeline before being classified into six intersectional classes by RF, GNB, LR, and SVM. Hyperparameter tuning is conducted via 5-Fold Stratified Cross-Validation with an isolated GridSearchCV scheme to prevent information leakage prior to test data evaluation.

A. DATASET

The dataset used in this study is DemogPairs [25], a dataset specifically designed for facial recognition across demographic groups. This dataset contains a total of 10,800 facial images distributed evenly across six intersectional classes. The six subgroups encompass cross-combinations of three macro-race groups and two gender groups, namely Asian Females, Asian Males, Black Females, Black Males, White Females, and White Males, with an exact quota of 1,800 images per class. This balanced distribution provides a more controlled evaluation condition to compare performance across

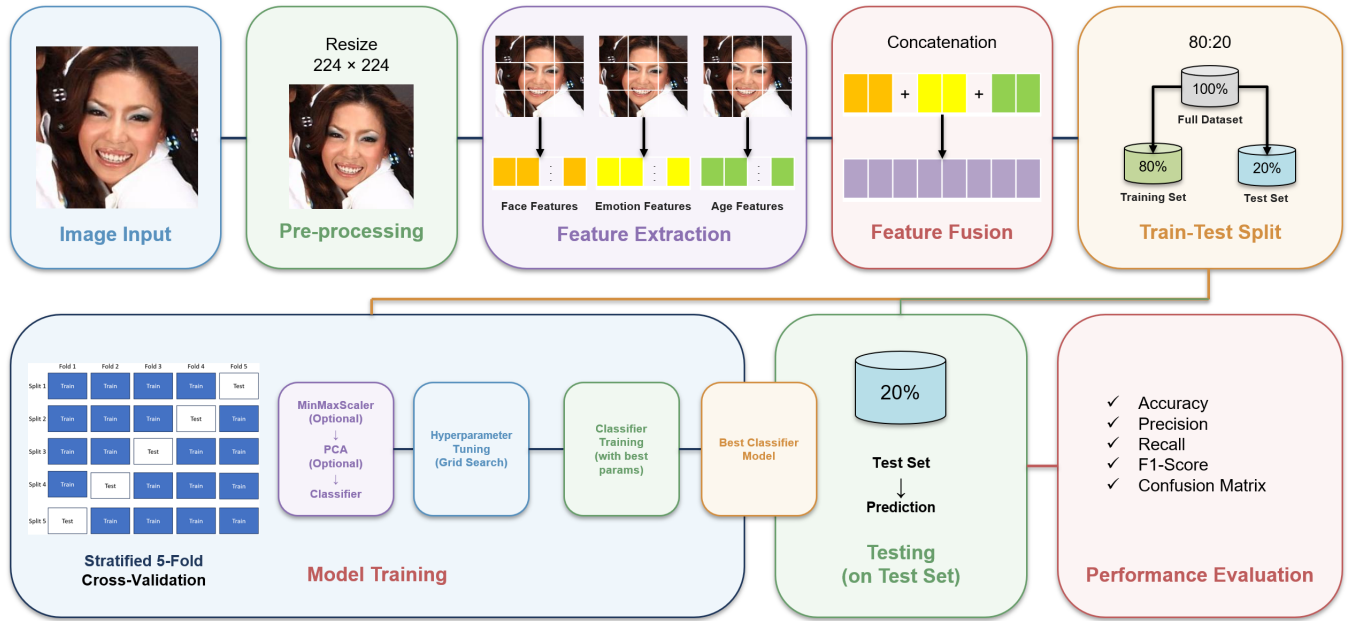


FIGURE 1. End-to-End Framework of the Proposed Multi-Domain Vision Transformer Feature Fusion and Classical Classifier Optimization for Intersectional Race and Gender Classification.



FIGURE 2. Sample Images of the DemogPairs Dataset across Six Intersectional Demographic Subgroups: (a) Asian Females, (b) Asian Males, (c) Black Females, (d) Black Males, (e) White Females, and (f) White Males.

subgroups by mitigating the influence of sample size imbalances [20]. Visual samples from the six demographic subgroups are illustrated in Figure 2.

TABLE 1. Dataset Partition and Demographic Subgroup Distribution.

Subgroup	Train Set (80%)	Test Set (20%)	Total
Black_Males	1,440	360	1,800
White_Females	1,440	360	1,800
Asian_Males	1,440	360	1,800
White_Males	1,440	360	1,800
Black_Females	1,440	360	1,800
Asian_Females	1,440	360	1,800
Total	8,640	2,160	10,800

To ensure the integrity of empirical testing, the dataset is partitioned using an 80/20 stratified split procedure (80% training data and 20% test data). This partition yields 8,640 training images with 1,440 samples per class and 2,160 independent test images with 360 samples per class, as detailed in Table 1. The test subset is strictly isolated, never involved during the hyperparameter search process, and solely used

for final evaluation. At the image standardization stage, each facial image is converted into a 3-channel Red, Green, Blue (RGB) format, resized to a resolution of 224×224 pixels, and its pixel intensity is scaled from the range $[0, 255]$ to $[0, 1]$ to align with the transformer model inputs.

B. VISION TRANSFORMER

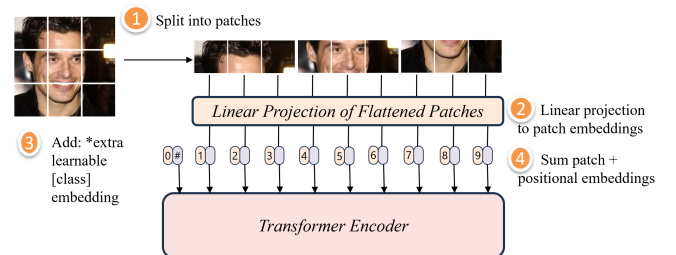


FIGURE 3. Architecture of the ViT Backbone and Patch Projection.

The visual representation extraction architecture employs the Vision Transformer (ViT) Base backbone, process-

ing two-dimensional facial images as sequences of one-dimensional patch tokens, as illustrated in Figure 3 [26]. An input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ with resolution $224 \times 224 \times 3$ is partitioned into $N = 196$ non-overlapping patches of size $P \times P = 16 \times 16$ pixels [27]. Each patch is linearly projected via matrix $\mathbf{E} \in \mathbb{R}^{(P^2 C) \times D}$ into a latent embedding space of dimension $D = 768$, and concatenated with a class token $\mathbf{x}_{\text{class}}$ and positional embeddings $\mathbf{E}_{\text{pos}}$ as formulated in (1) [27]. The token sequence propagates through $L = 12$ transformer encoder layers, each comprising a Multi-Head Self-Attention (MHSA) module and an MLP block with Layer Normalization (LN) as in (2) and (3) [28]. The self-attention mechanism captures global spatial dependencies across all image patches without receptive field constraints inherent to conventional convolutions [29].

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \dots; \mathbf{x}_p^{196} \mathbf{E}] + \mathbf{E}_{\text{pos}} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MHSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1} \quad (2)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell \quad (3)$$

TABLE 2. Multi-Domain Feature Fusion and Ablation Configurations.

#	Configuration	Domain Category	Dimension
1	Face	Single-Domain	768
2	Emotion	Single-Domain	768
3	Age	Single-Domain	768
4	Emotion $\oplus$ Face	Dual-Domain	1,536
5	Face $\oplus$ Age	Dual-Domain	1,536
6	Emotion $\oplus$ Age	Dual-Domain	1,536
7	Face $\oplus$ Emotion $\oplus$ Age	Tri-Domain (Proposed)	2,304

To ensure computational efficiency, the three ViT backbones remain frozen as offline feature extractors. The checkpoints comprise ViT-Face ('skutaada/VIT-VGGFace') capturing facial biometric geometry, ViT-Emotion ('dima806/facial_emotions_image_detection') capturing affective dynamic expressions, and ViT-Age ('dima806/facial_age_image_detection') capturing biological aging morphology. For each model, latent feature representation $\mathbf{f}_{\text{domain}}$ is extracted from the [CLS] token of final encoder layer L via Layer Normalization in (4). Multi-domain feature fusion integrates extracted latent vectors via direct sum concatenation, producing a 2,304-dimensional tri-domain vector $\mathbf{z}_{\text{tri}}$ in (5). To systematically evaluate individual and joint domain contributions, seven feature representation ablation configurations are formulated as detailed in Table 2.

$$\mathbf{f}_{\text{domain}} = \text{LN}(\mathbf{z}_L^0) \in \mathbb{R}^{768} \quad (4)$$

$$\mathbf{z}_{\text{tri}} = \mathbf{f}_{\text{face}} \oplus \mathbf{f}_{\text{emotion}} \oplus \mathbf{f}_{\text{age}} \in \mathbb{R}^{2304} \quad (5)$$

C. RANDOM FOREST

The RF ensemble aggregates predictions from independent decision trees to classify facial features using bootstrap aggregating (bagging) and random subspaces [30]. This approach reduces prediction variance by constructing B classification trees from random samples of the training data [31]. Each node split restricts feature selection to a random subset of size $\text{max_features} \in \{\text{'sqrt'}, \text{'log2'}\}$ to suppress inter-tree correlation [32]. The split criterion optimizes the Gini impurity index until satisfying the minimum node split threshold min_samples_split , the maximum depth limit max_depth , or the minimum leaf threshold min_samples_leaf [33]. The majority voting mechanism determines the final decision for an image sample across all decision trees [34]. Hyperparameter tuning evaluates feature scaling, PCA dimensionality reduction, tree counts n_estimators , and tree growth constraints, producing 288 grid configurations with 1,440 total fits via 5-Fold Stratified Cross-Validation in Table 3.

D. GAUSSIAN NAIVE BAYES

Under Bayes' theorem, the GNB classifier estimates class posterior probabilities assuming continuous feature dimensions are conditionally independent [35]. A Gaussian probability density function models feature likelihood per target class in (6), where x_i is the i -th feature, y is the class label, c is the target class, and $\mu_{c,i}$ and $\sigma_{c,i}^2$ are the mean and variance in class c [35]:

$$P(x_i | y = c) = \frac{1}{\sqrt{2\pi\sigma_{c,i}^2}} \exp\left(-\frac{(x_i - \mu_{c,i})^2}{2\sigma_{c,i}^2}\right) \quad (6)$$

The smoothing parameter var_smoothing maintains numerical stability against near-zero variance in high-dimensional latent representation spaces [36]. The hyperparameter tuning procedure via 5-Fold Stratified Cross-Validation evaluates combinations of data scaling, PCA dimensionality reduction, and 40 var_smoothing intervals explored logarithmically from 1.0×10^{-9} to 1.0×10^2 , yielding 240 grid combinations with a total of 1,200 fits as presented in Table 4.

E. LOGISTIC REGRESSION

Multinomial LR (Softmax regression) maps latent representations into six intersectional demographic classes via normalized posterior probabilities [37]. For input vector $\mathbf{x} \in \mathbb{R}^D$ and class $c \in \{1, \dots, K\}$ with $K = 6$, the conditional probability is formulated in (7), where $\mathbf{w}_c$ and b_c denote the weight vector and bias parameter of class c [38]:

$$P(y = c | \mathbf{x}) = \frac{e^{\mathbf{w}_c^T \mathbf{x} + b_c}}{\sum_{j=1}^K e^{\mathbf{w}_j^T \mathbf{x} + b_j}} \quad (7)$$

Model training minimizes categorical cross-entropy loss regularized with an L_2 penalty to prevent overfitting on high-dimensional representations [39]. Hyperparameter tuning via 5-Fold Stratified Cross-Validation evaluates variations

TABLE 3. Hyperparameter Search Space for Random Forest Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Number of Estimators	100, 200	2
Maximum Tree Depth	None, 20, 30	3
Feature Subspace	'sqrt', 'log2'	2
Minimum Samples Split	2, 5	2
Minimum Samples Leaf	1, 2	2
Total Grid Combinations		288 (1,440 fits)

TABLE 4. Hyperparameter Search Space for Gaussian Naive Bayes Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Variance Smoothing	logspace($-9, 2, 40$) (1.0×10^{-9} to 1.0×10^2)	40
Total Grid Combinations		240 (1,200 fits)

TABLE 5. Hyperparameter Search Space for Logistic Regression Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Regularization Strength	0.01, 0.1, 1, 10, 100	5
Optimization Solver	'lbfgs', 'saga', 'newton-cg'	3
Maximum Iterations	500, 1000, 2000	3
Total Grid Combinations		270 (1,350 fits)

of feature scaling, PCA dimensionality reduction, five regularization values $C \in \{0.01, 0.1, 1, 10, 100\}$, three solvers ('lbfgs', 'saga', 'newton-cg'), and three maximum iteration limits (200, 500, 2000), producing 270 grid combinations with 1,350 total fits in Table 5.

F. SUPPORT VECTOR MACHINE

The SVM classifier constructs a maximum margin hyperplane to separate the six intersectional classes in the high-dimensional latent representation space [40]. Kernel functions project the nonlinear data distribution from multi-domain fusion into a higher-dimensional feature space, encompassing linear, Radial Basis Function (RBF), and polynomial kernel options [41]. The second-degree polynomial kernel formulation in (8) computes the dot product $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ between pairs of feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, where γ denotes the kernel scaling parameter, $d = 2$ determines the mapping degree, and the intercept constant $\text{coef0} = 0.0$ follows the scikit-learn default value [42]:

$$K(\mathbf{x}_i, \mathbf{x}_j) = (\gamma \langle \mathbf{x}_i, \mathbf{x}_j \rangle + \text{coef0})^d, \quad d = 2 \quad (8)$$

This quadratic mapping facilitates the efficient capture of nonlinear interactions across visual dimensions [43].

Hyperparameter optimization evaluates penalty parameter ranges $C \in \{0.01, 0.1, 1, 10\}$, various kernel functions, γ coefficients, polynomial degrees, as well as scaling and PCA

options, yielding 288 configuration combinations or 1,440 total fits across the 5-Fold Stratified Cross-Validation procedure as presented in Table 6.

G. CLASSIFICATION PIPELINE

The modular pipeline framework integrates feature scaling, representational dimensionality reduction, and downstream estimation into a unified transformation chain formulated in (9), where input feature vector $\mathbf{x}$ is scaled to $\tilde{\mathbf{x}}$, reduced via PCA to $\hat{\mathbf{x}}$, and mapped by the classifier to predicted label $\hat{y} \in \{0, 1, 2, 3, 4, 5\}$ for the six demographic subgroups:

$$\mathbf{x} \xrightarrow{\text{Scaler}} \tilde{\mathbf{x}} \xrightarrow{\text{PCA}} \hat{\mathbf{x}} \xrightarrow{\text{Classifier}} \hat{y} \in \{0, 1, 2, 3, 4, 5\} \quad (9)$$

The cross-validation protocol implements a 5-Fold Stratified Cross-Validation scheme with controlled shuffling to maintain a balanced proportion of $\sim 16.67\%$ per class in each fold. To prevent information leakage, the Scaler and PCA are fitted exclusively on training data within GridSearchCV before transform application. Evaluating 1,086 hyperparameter combinations yields 5,430 fits per scheme or 38,010 total fits across the seven configurations, along with 28 final refits before held-out testing.

H. EVALUATION METRICS

Multi-class classification performance across six demographic subgroups is evaluated at the subgroup level using a

TABLE 6. Hyperparameter Search Space for Support Vector Machine.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Regularization Parameter	0.01, 0.1, 1, 10	4
Kernel Function	'linear', 'rbf', 'poly'	3
Polynomial Degree	2, 3	2
Kernel Coefficient	'scale', 'auto'	2
Total Grid Combinations		288 (1,440 fits)

binary One-vs-Rest (OvR) scheme [44]. This scheme utilizes four per-class confusion matrix components to compute OvR Accuracy, Precision, Recall, and F1-Score for each class $c \in \{1, \dots, K\}$ [45]. These components comprise TP_c (True Positive), TN_c (True Negative), FP_c (False Positive), and FN_c (False Negative) [46]. OvR Accuracy in (10) measures the ratio of total correct predictions to the test data, whereas Precision in (11) and Recall in (12) map positive precision and detection sensitivity per subgroup [47]. The F1-Score in (13) is the harmonic mean of Precision and Recall for balanced subgroup evaluation [47].

$$\text{Accuracy}_c = \frac{TP_c + TN_c}{TP_c + TN_c + FP_c + FN_c} \quad (10)$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (11)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (12)$$

$$\text{F1-Score}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (13)$$

To evaluate overall classification performance on held-out test data, subgroup metrics are aggregated into global performance metrics. In this formulation, N denotes total test samples ($N = 2,160$), while K represents the total demographic subgroups ($K = 6$). Global Accuracy (Overall Accuracy) in (14) is calculated as the ratio of total correct predictions, namely the sum of diagonal confusion matrix elements $\sum_{c=1}^K TP_c$, to total test samples N .

$$\text{Accuracy}_{\text{Global}} = \frac{\sum_{c=1}^K TP_c}{N} \quad (14)$$

Subsequently, global Precision (Macro Precision) in (15), global Recall (Macro Recall) in (16), and global F1-Score (Macro F1-Score) in (17) are computed via an unweighted macro average across $K = 6$ subgroups to assign equal evaluation weight to each demographic class.

$$\text{Precision}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{Precision}_c \quad (15)$$

$$\text{Recall}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{Recall}_c \quad (16)$$

$$\text{F1-Score}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{F1-Score}_c \quad (17)$$

IV. RESULTS AND DISCUSSION

A. GLOBAL PERFORMANCE ANALYSIS

Performance evaluation of RF across seven feature configurations on the DemogPairs test data is presented in Table 7. Experimental results demonstrate that the dual-domain Emotion $\oplus$ Face scheme achieves peak performance, recording an Accuracy of 86.85% and an F1-Score of 86.82%. In contrast, tri-domain Face $\oplus$ Emotion $\oplus$ Age exhibits slight degradation to an Accuracy of 86.20% and an F1-Score of 86.13%, reflecting challenges in partitioning 2,304 dimensions via random tree splits [48]. All GridSearchCV configurations select PCA 0.75 dimensionality reduction to balance data variance. In the single-domain category, Face achieves the best performance with an Accuracy of 85.46%, surpassing Emotion at 80.60% and Age at 73.66%.

The performance evaluation of probabilistic GNB classification across seven feature configurations is summarized in Table 8. The GNB model demonstrates a consistent performance improvement from single-domain to multi-domain categories. In the single-domain category, the Age configuration yields the lowest performance with an Accuracy of 69.63% and an F1-Score of 69.52%, whereas Face achieves an Accuracy of 82.69%. Feature integration in dual-domain schemes consistently enhances classification accuracy, led by Emotion $\oplus$ Face at 84.86%. Peak GNB classification performance is attained by tri-domain Face $\oplus$ Emotion $\oplus$ Age with 85.05% Accuracy and 85.05% F1-Score. All GNB configurations uniformly select PCA 0.75 dimensionality reduction via GridSearchCV across all feature schemes.

The performance benchmark results of multinomial LR across all feature configurations are summarized in Table 9. A key characteristic of the LR model is full feature retention, wherein all seven schemes select no PCA reduction ($\text{pca}=\text{None}$) [49]. Empirical evaluation demonstrates that the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age achieves the highest performance with 92.73% Accuracy and 92.73% F1-Score using penalty parameter $C = 0.1$ and the newton-cg solver. This tri-domain achievement surpasses the best dual-domain configuration Emotion $\oplus$ Face (92.41% Accuracy, 92.40% F1-Score) and single-domain Face (90.60% Accuracy). These findings indicate the effectiveness of L_2 regularization in handling high-dimensional feature spaces without principal component compression [50].

(Note: The degree parameter is active and reported only for the poly kernel; for the rbf kernel, the degree parameter is

TABLE 7. Performance Benchmark of Random Forest across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	85.46%	85.43%	85.46%	85.39%	n_est=200, depth=30, max_feat=log2, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=MinMaxScaler
Emotion	80.60%	80.63%	80.60%	80.57%	n_est=200, depth=None, max_feat=log2, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None
Age	73.66%	73.63%	73.66%	73.54%	n_est=200, depth=30, max_feat=log2, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=None
Emotion $\oplus$ Face	86.85%	86.89%	86.85%	86.82%	n_est=200, depth=None, max_feat=sqrt, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None
Face $\oplus$ Age	85.79%	85.78%	85.79%	85.73%	n_est=200, depth=None, max_feat=sqrt, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=None
Emotion $\oplus$ Age	81.11%	81.11%	81.11%	81.08%	n_est=200, depth=None, max_feat=log2, min_split=5, min_leaf=2, pca=PCA(0.75), scaler=None
Face $\oplus$ Emotion $\oplus$ Age	86.20%	86.20%	86.20%	86.13%	n_est=200, depth=30, max_feat=sqrt, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None

TABLE 8. Performance Benchmark of Gaussian Naive Bayes across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	82.69%	82.71%	82.69%	82.58%	var_smoothing=4.1246e-02, pca=PCA(0.75), scaler=MinMaxScaler
Emotion	73.38%	73.87%	73.38%	73.29%	var_smoothing=3.0703e-03, pca=PCA(0.75), scaler=None
Age	69.63%	69.79%	69.63%	69.52%	var_smoothing=4.3755e-04, pca=PCA(0.75), scaler=MinMaxScaler
Emotion $\oplus$ Face	84.86%	84.90%	84.86%	84.81%	var_smoothing=5.8780e-03, pca=PCA(0.75), scaler=MinMaxScaler
Face $\oplus$ Age	83.15%	83.43%	83.15%	83.17%	var_smoothing=1.1253e-02, pca=PCA(0.75), scaler=MinMaxScaler
Emotion $\oplus$ Age	76.81%	76.86%	76.81%	76.81%	var_smoothing=1.6037e-03, pca=PCA(0.75), scaler=MinMaxScaler
Face $\oplus$ Emotion $\oplus$ Age	85.05%	85.12%	85.05%	85.05%	var_smoothing=5.8780e-03, pca=PCA(0.75), scaler=None

inactive and omitted from the publication manuscript).

The performance progression of SVM across all feature ablation stages is summarized in Table 10. Similar to the linear model, all SVM configurations consistently maintain the full representation space without PCA reduction (pca=None). Empirical evaluation reveals a tiered performance increase from single-domain Face (90.83% Accuracy, 90.83% F1-Score) to dual-domain Emotion $\oplus$ Face (93.29% Accuracy, 93.29% F1-Score), peaking in tri-domain Face $\oplus$ Emotion $\oplus$ Age with an Accuracy of 93.70% and an F1-Score of 93.69%. In terms of hyperparameters, the degree parameter value of 2 is active only for the poly kernel function, whereas RBF kernel-based models do not activate this parameter within the evaluated search space [51].

The results across the 28 experiments in Tables 7–10 demonstrate that SVM with tri-domain fusion Face $\oplus$ Emotion $\oplus$ Age delivers the highest performance, achieving 93.70% Accuracy and 93.69% F1-Score. Averaged across the seven feature configurations, the overall classifier performance hierarchy follows: SVM (91.47%) > LR (90.40%) > RF (82.81%) > GNB (79.37%). For the tri-domain SVM configuration, GridSearchCV selects a degree-2 polynomial kernel as the optimal configuration for demographic classification. This empirical finding applies specifically to the DemogPairs dataset and evaluated search space. This com-

parative evaluation remains descriptive without claiming statistical superiority, as this study does not conduct formal statistical hypothesis testing across classifiers.

B. FEATURE ABLATION STUDY

The quantification of classification accuracy changes from single-domain schemes toward dual- and tri-domain configurations is summarized in Table 7, Table 8, Table 9, and Table 10. In the SVM model, the representation transition from Face (90.83%) to Emotion $\oplus$ Face (93.29%) increases accuracy by 2.46%. The addition of a third domain in the tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration yields a further performance improvement to 93.70%, which reflects a cumulative increase of 2.87% over the single Face feature. This pattern of progressive improvement indicates the presence of additional discriminative information from each combined visual representation domain, although the interaction mechanisms among latent feature dimensions cannot be definitively confirmed solely from empirical test results.

An analysis of the contribution of the age domain (Age) representation dissects the role of latent aging information within the demographic classification system. Across all evaluated classifiers, Age constitutes the single-domain configuration with the lowest accuracy attainment, as recorded at 87.64% for SVM and 69.63% for GNB. Nevertheless,

TABLE 9. Performance Benchmark of Logistic Regression across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	90.60%	90.60%	90.60%	90.59%	$C = 1$, solver=newton-cg, max_iter=500, pca=None, scaler=MinMaxScaler
Emotion	88.47%	88.50%	88.47%	88.46%	$C = 1$, solver=saga, max_iter=500, pca=None, scaler=MinMaxScaler
Age	86.48%	86.49%	86.48%	86.48%	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Emotion $\oplus$ Face	92.41%	92.41%	92.41%	92.40%	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Face $\oplus$ Age	91.62%	91.62%	91.62%	91.62%	$C = 0.1$, solver=newton-cg, max_iter=500, pca=None, scaler=None
Emotion $\oplus$ Age	90.51%	90.52%	90.51%	90.51%	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Face $\oplus$ Emotion $\oplus$ Age	92.73%	92.75%	92.73%	92.73%	$C = 0.1$, solver=newton-cg, max_iter=500, pca=None, scaler=None

TABLE 10. Performance Benchmark of Support Vector Machine across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	90.83%	90.84%	90.83%	90.83%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Emotion	90.19%	90.20%	90.19%	90.17%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Age	87.64%	87.67%	87.64%	87.65%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Emotion $\oplus$ Face	93.29%	93.33%	93.29%	93.29%	$C = 10$, rbf, γ =scale, pca=None, scaler=MinMaxScaler
Face $\oplus$ Age	92.55%	92.54%	92.55%	92.54%	$C = 10$, poly, γ =scale, deg=2, pca=None, scaler=None
Emotion $\oplus$ Age	92.08%	92.10%	92.08%	92.09%	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Face $\oplus$ Emotion $\oplus$ Age	93.70%	93.72%	93.70%	93.69%	$C = 10$, poly, γ =scale, deg=2, pca=None, scaler=None

combining the Age feature with facial biometric features in the dual-domain Face $\oplus$ Age scheme yields an accuracy of 92.55%, providing an improvement of 1.72% over the pure Face feature performance (90.83%) on the SVM model. These empirical results indicate the likely presence of additional discriminative information from age-associated representations that complements the facial morphological profile, without claiming that such informational contribution is orthogonal or strictly complementary theoretically.

C. INTERSECTIONAL SUBGROUP PERFORMANCE

The granular classification performance profile of the best-performing tri-domain SVM model (Face $\oplus$ Emotion $\oplus$ Age) across the six demographic subgroups is thoroughly evaluated via OvR metrics in Table 11. All demographic subgroups successfully achieve an F1-Score exceeding 91.00%, ranging from 91.74% for Black Females to 96.14% for White Males, resulting in a 4.40% disparity gap. Concurrently, OvR Accuracy values fall within the range of 97.31% to 98.70% with a 1.39 pp disparity. Methodologically, high OvR Accuracy values are influenced by the 5:1 negative sample dominance in binary OvR evaluation, such that this metric is not positioned as the sole indicator of performance disparity across subgroups.

A comparison of subgroup performance profiles between

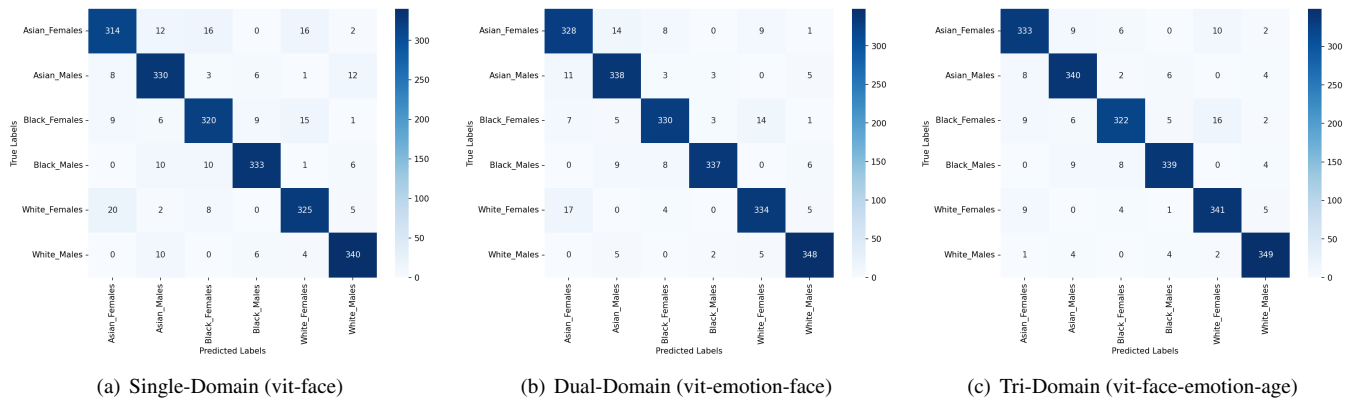
the SVM model and the linear LR classifier under the tri-domain configuration is presented in Table 11. The tri-domain LR model likewise demonstrates high performance consistency, with F1-Score achievements exceeding 91.00% across all demographic classes, spanning from 91.36% for Asian Females to 95.58% for White Males with a range disparity difference of 4.22%. The White Males subgroup consistently records the highest classification performance across both classifiers. These empirical findings demonstrate that the tri-domain representation fusion yields robust visual representations that are stably distributed across all demographic subgroups, in both linear and non-linear models. Nevertheless, this subgroup evaluation reflects granular per-class performance rather than a comprehensive algorithmic fairness audit.

D. ERROR PATTERN ASSESSMENT

The shift in classification error distributions from the single-domain scheme toward dual- and tri-domain configurations is analyzed through the confusion matrix diagrams in Figure 4, which encompass Figure 4(a), Figure 4(b), and Figure 4(c). Empirical evaluation reveals that the total misclassifications across the 2,160 held-out test data consistently decrease as the latent representation domains expand. The single-domain Face SVM model yields 198 prediction errors, which subse-

TABLE 11. Subgroup-Level Classification Performance across Six Intersectional Demographics for Top-Performing Tri-Domain Models.

Classifier	Subgroup	Recall	Precision	F1-Score	OvR Accuracy
SVM (Tri)	White_Males	96.94%	95.36%	96.14%	98.70%
	Black_Males	94.17%	95.49%	94.83%	98.29%
	White_Females	94.72%	92.41%	93.55%	97.82%
	Asian_Males	94.44%	92.39%	93.41%	97.78%
	Asian_Females	92.50%	92.50%	92.50%	97.50%
	Black_Females	89.44%	94.15%	91.74%	97.31%
LR (Tri)	White_Males	96.11%	95.05%	95.58%	98.52%
	Black_Males	93.06%	95.71%	94.37%	98.15%
	White_Females	92.22%	91.21%	91.71%	97.22%
	Asian_Males	92.78%	90.76%	91.76%	97.22%
	Asian_Females	91.11%	91.62%	91.36%	97.13%
	Black_Females	91.11%	92.13%	91.62%	97.22%

**FIGURE 4. Normalized Confusion Matrices of Selected Support Vector Machine Models on the Held-Out Test Set (N = 2,160): (a) Single-Domain (vit-face), (b) Dual-Domain (vit-emotion-face), and (c) Tri-Domain (vit-face-emotion-age).**

quently decrease to 145 samples in the dual-domain Emotion $\oplus$ Face scheme. The tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration suppresses classification errors to the lowest point of 136 samples. This consistent reduction in errors indicates that multi-domain representation fusion sharpens the separation boundaries between demographic subgroups in the latent feature space.

An in-depth examination of the confusion matrix for the single-domain Face model in Figure 4(a) reveals that classification errors are predominantly concentrated in inter-racial subgroups within the same gender. This tendency is particularly pronounced in the female cohort, where the Asian Females subgroup incurs 46 misclassifications, encompassing 16 samples predicted as White Females and 16 samples as Black Females. Conversely, the White Females subgroup records 20 prediction errors distributed to Asian Females. This empirical pattern is consistent with the potential presence of inter-racial phenotypic overlap within single facial biometric geometric representations [18]. Nevertheless, such confusion matrix evidence does not prove that visual overlap is the sole cause of all observed misclassifications.

The shift in classification error distributions in the dual-domain Emotion $\oplus$ Face model is further analyzed based on the confusion matrix in Figure 4(b). The addition of affective expression latent representations effectively reduces the inter-

racial misclassifications among same-gender subgroups, particularly within the female group. Prediction errors from Asian Females to White Females are substantially suppressed from 16 cases to 9 cases. Concurrently, correctly predicted samples (true positives) in the White Females subgroup increase from 325 images to 334 images. Integrating dynamic facial expression signals strengthens inter-racial visual discriminant boundaries, enabling the classifier to distinguish similar morphological features with a more robust and stable certainty.

The misclassification patterns in the tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration are comprehensively evaluated based on the confusion matrix in Figure 4(c). The integration of age-domain representations consistently increases the number of correct predictions in the Asian Females subgroup to 333 samples and White Females to 341 samples. Furthermore, cross-gender misclassifications, representing instances where female subjects are confused as male or vice versa, amount to only 51 cases out of the 2,160 held-out test data (2.36%). These empirical findings indicate that inter-gender boundaries are separated very distinctly within the multi-domain latent representation space, such that the remaining classification errors in the tri-domain model are almost entirely dominated by inter-racial misclassifications within the same gender group.

E. COMPARISON WITH PRIOR STUDIES

TABLE 12. Performance Comparison with Prior Intersectional Studies on the DemogPairs Benchmark.

Model	Accuracy	Precision	Recall	F1-Score
MD-ViT [20]	89.07%	89.12%	89.07%	89.01%
Dual-ViT [19]	92.41%	92.48%	92.41%	92.38%
Ours	93.70%	93.72%	93.70%	93.69%

A direct empirical comparison between the proposed framework and prior studies evaluated on the DemogPairs dataset is summarized in Table 12. The proposed model based on Tri-Domain ViT + SVM (Face $\oplus$ Emotion $\oplus$ Age) achieves an Accuracy of 93.70%, a Precision of 93.72%, a Recall of 93.70%, and an F1-Score of 93.69%. These quantitative achievements surpass the metrics reported for the Dual-ViT + SVM model (92.41% Accuracy, 92.38% F1-Score) and MD-ViT + XGBoost (89.07% Accuracy, 89.01% F1-Score) [19], [20]. All comparative figures are directly cited from respective publications, serving as contextual positioning on the benchmark dataset rather than an identical experimental replication.

The seminal research on DemogPairs established an essential foundation for quantifying the impact of demographic imbalance on facial recognition systems [25]. In response to this challenge, the three-domain representation integration framework proposed in this study achieves the highest accuracy among compared studies on the DemogPairs dataset [19], [20]. In addition to evaluating global aggregate metrics, this investigation conducts a comprehensive subgroup disparity analysis on the best tri-domain configuration to transparently map performance variations across demographic subgroups. This detailed evaluation confirms recognition stability across intersectional race and gender categories, though cross-scheme disparity comparison with prior studies was precluded by unrecorded literature metrics [19], [20].

V. CONCLUSION

This study evaluates a Tri-Domain ViT latent representation fusion framework (Face $\oplus$ Emotion $\oplus$ Age: 2,304 dimensions) for intersectional race and gender classification on the DemogPairs dataset. Experimental results demonstrate that tri-domain fusion achieves the best configuration across 3 of the 4 evaluated classifiers, encompassing SVM, LR, and GNB. Within the evaluated search space, the optimized SVM model with hyperparameter $C = 10$ and a degree-two polynomial kernel yields the highest classification performance, namely an Accuracy of 93.70% and an F1-Score of 93.69%. Nevertheless, the benefit of multi-domain representation integration proves to be classifier-dependent, as RF achieves its peak performance on the dual-domain Emotion $\oplus$ Face scheme with an accuracy of 86.85%. In addition, subgroup performance evaluation confirms that the F1-Score values across all six demographic subgroups are stably distributed within the range of 91.74% to 96.14%.

Despite these achievements, this study has several limitations that open avenues for future research. First, the demographic scope of this investigation is limited to three macro-racial groups (Asian, Black, and White) in the DemogPairs dataset, so the obtained empirical findings cannot be directly generalized to multiracial populations or other ethnic groups without additional comprehensive testing. Second, latent representation extraction is conducted separately (offline) utilizing frozen pre-trained ViT backbones without simultaneous end-to-end fine-tuning. Third, the experimental evaluation remains focused on controlled laboratory environment images, such that the model's response to extreme pose variations, occlusions, and unconstrained (in-the-wild) illumination requires further investigation. Therefore, future research directions are recommended to explore cross-modal adaptive attention fusion mechanisms (cross-attention transformer fusion) trained end-to-end to learn dynamic inter-domain interaction weights, as well as extending evaluation to larger, more heterogeneous datasets such as FairFace or UTKFace. In addition, the implementation of visual explainability methods such as Attention Rollout or Grad-CAM is crucial to transparently map the spatial contribution of each domain, accompanied by the enforcement of ethical artificial intelligence governance (Responsible AI) with human-in-the-loop oversight to prevent exploitation in public surveillance systems.

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